

Control Systems A

Week 1 - Introduction and Control System Components

UoN - FEE 3411

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What we are doing today

- **Hour 1 — ideas.** What a control system is, what its parts are called, and why feedback is worth its cost. Almost no hardware.
- **Hour 2 — hardware.** A survey of the devices that fill the boxes in a block diagram. Almost no mathematics.

Why the second hour matters

For the next eleven weeks a block on a diagram will be labelled “actuator”. This hour is so that you know what is physically inside it.

Syllabus: *Introduction: Examples of control systems. Open and closed-loop classification. Control systems components and devices. Servomechanism, Stepper motors, Hydraulic systems, Pneumatic systems, Valves, cylinders, and rotary actuators.* ELO 1.

What control engineering is

What do these have in common?

- Rockets and aircraft
- Cars and autonomous vehicles
- Robots and machine tools
- Power grids and power stations
- Electric motor drives
- Manufacturing processes
- Disc drives
- Artificial pancreas systems

Key idea

Every one contains an **automatic control system**: an arrangement that makes some variable behave in a desired way *without a human operator intervening moment by moment*.

Khalil, p. 2.

Concretely, “behave in a desired way” means

- hold a car at a set speed, or keep it inside a lane;
- hold a room at a set temperature;
- make the position or velocity of a motor follow a prescribed pattern;
- hold the frequency of a power grid at 50 Hz;
- make a rocket follow a desired trajectory.

All of these share one skeleton:

- a physical thing whose behaviour we care about;
- variables we can **measure** — outputs;
- variables that **drive** it — inputs.

Key idea

A control problem exists whenever you can measure something, you can push on something, and there is a gap between what the thing is doing and what you want it to do.

Two kinds of input

Control inputs

Inputs we **can** manipulate.

Throttle angle, armature voltage, valve opening, furnace on/off time.

Disturbance inputs

Inputs that affect the plant but are **beyond our control**.

Road gradient, wind load, a change in load torque, heat loss to the outside.

- In heating a house we control when the furnace fires; we do not control the heat lost through the walls.
- Throughout this unit: **one** control input, **one** measured output — *single-input-single-output* (SISO).

The vocabulary — fixed now for twelve weeks

Term	Meaning
Plant (process)	The system whose output is to be controlled.
Controlled variable	The output we want to behave in a desired way.
Reference	The value we want the controlled variable to take. Also: command, set point, desired output.
Control input	An input we can manipulate.
Disturbance input	An input we cannot manipulate.

Learn these this week. From next week they are assumed.

The vocabulary (continued)

Term	Meaning
Actuator	Converts the controller's low-power command into the physical quantity acting on the plant.
Sensor	Measures the controlled variable and converts it to the form the controller uses.
Measurement noise	The difference between what the sensor reports and what is actually there.
Controller	Computes the control input from the reference and the measurement.
Error	Reference minus controlled variable — exactly, only when both transducers have unit gain.

A warning about the word “input”

Common pitfall

“Input” is ambiguous in ordinary speech and precise here.

- The **input to the system** is the reference $r(t)$ — what you ask for.
- The **input to the plant** is the control signal $u(t)$ — what the actuator delivers.

They are different signals at different points of the diagram. Confusing them is the single most common cause of a wrong block-diagram reduction in Week 4.

Open-loop and closed-loop control

Two architectures, and only two

Open-loop control

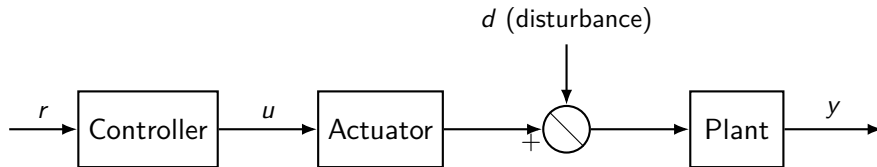
A model of the plant is used to work out, *in advance*, what control input ought to produce the desired output. That input is applied. Nothing is measured, so nothing is corrected.

Closed-loop (feedback) control

The output is *measured*, fed back, and compared with the reference. The control input is computed from the difference, in a direction that drives the output towards the reference.

Khalil, p. 2; Nise §1.3, p. 7.

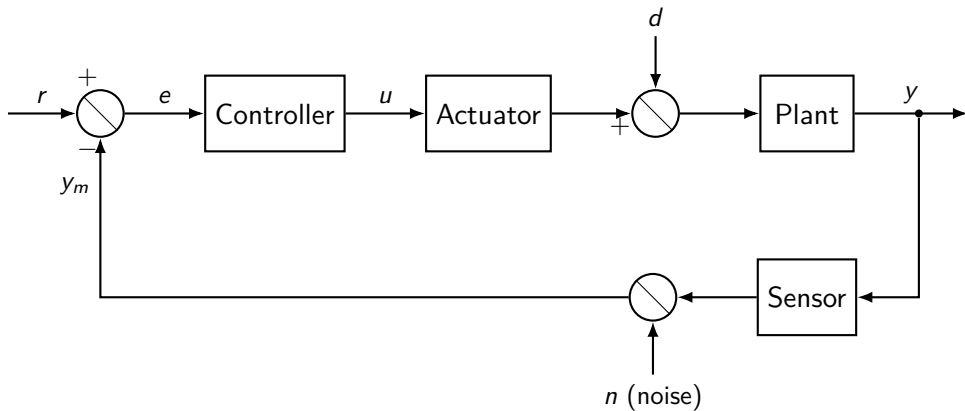
Open loop — nothing is measured



The defining weakness

The disturbance d passes straight through to the output and is **never corrected**. The system is simply commanded by its input.

Closed loop — the output is measured and fed back



The disturbance now shows up in the measurement, hence in the error e , so the controller acts on it. The price is the sensor: its cost, and its noise n . Cf. Nise Fig. 1.6, p. 8.

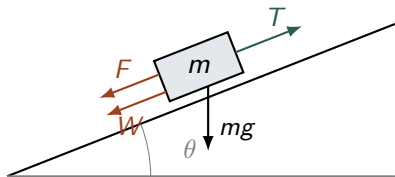
The toaster

- A toaster's output is the **colour** of the toast.
- It measures **time**.
- So it cannot know that the bread is rye rather than white, or thick rather than thin.
- Anyone who has burnt toast has performed the experiment.
- A toaster *oven* that measures reflectivity and humidity **is** closed-loop — and costs several times as much.

Key idea

The distinguishing feature of an open-loop system is that it cannot compensate for a disturbance. A closed-loop system corrects for disturbances *it can see in the measurement* — and only for those.

Worked example: cruise control — the model



$T = au$ thrust from throttle angle u , engine gain a

$F = bv$ road friction

$W = mg \sin \theta + D$ gradient + air drag

Newton's second law along the road:

$$m\dot{v} = au - bv - mg \sin \theta - D$$

What we know: the engine gain a .

What we don't: m (passengers), b (road surface), W (gradient, wind) — and b , W may drift with time.

Cf. Khalil Fig. 1-1, p. 3.

Cruise control — the open-loop design

Design as if there were no disturbance, $W = 0$, using *nominal* values $\hat{m}$, $\hat{b}$:

$$\hat{m}\dot{v} = au - \hat{b}v \quad \Longrightarrow \quad v_{ss} = \frac{au}{\hat{b}} \quad \text{at } \dot{v} = 0.$$

To make $v_{ss} = v_{des}$, choose the constant throttle

$$u = \frac{\hat{b}v_{des}}{a}.$$

Now apply that fixed u to the *real* car — true b , true m , real disturbance W :

$$m\dot{v} = \hat{b}v_{des} - bv - W \quad \Longrightarrow \quad v_{ss} = \frac{\hat{b}v_{des} - W}{b}$$

Open-loop steady-state error

$$v_{des} - v_{ss} = \left(\frac{b - \hat{b}}{b} \right) v_{des} + \frac{1}{b} W \quad (1)$$

Cruise control — the closed-loop design

Measure v with a speedometer and set

$$u = \underbrace{\frac{\hat{b} v_{des}}{a}}_{\text{feedforward}} + \underbrace{K(v_{des} - v)}_{\text{feedback}}$$

Substituting into the true model,

$$m\dot{v} = -(b + aK)v + (\hat{b} + aK)v_{des} - W$$

Closed-loop steady-state error

$$v_{des} - v_{ss} = \left(\frac{b - \hat{b}}{b + aK} \right) v_{des} + \frac{1}{b + aK} W \quad (3)$$

Cruise control — what it means

$$(1) \quad \left(\frac{b - \hat{b}}{b} \right) v_{des} + \frac{1}{b} W$$

$$(3) \quad \left(\frac{b - \hat{b}}{b + aK} \right) v_{des} + \frac{1}{b + aK} W$$

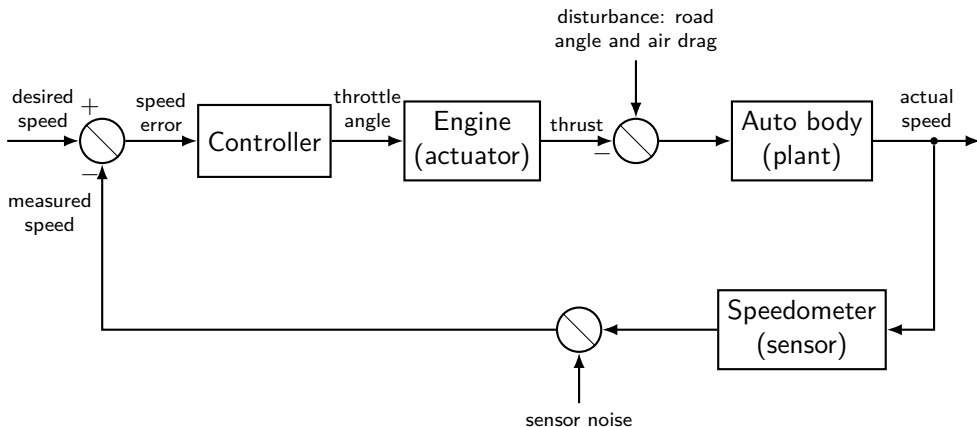
Key idea

The two expressions are identical except that b has been replaced by $b + aK$ in **both** denominators.

Raising the feedback gain K therefore shrinks **both** error terms at once: the one caused by not knowing b , and the one caused by the disturbance W .

One knob buys robustness to parameter error and disturbance rejection together. That is what feedback is for.

The cruise-control loop



The four basic components: **plant**, **actuator**, **sensor**, **controller**. Cf. Khalil Fig. 1-2, p. 5.

So why not simply make K enormous?

- 1 **The actuator saturates.** A throttle angle has a maximum. Beyond it, extra commanded control does nothing.

Control constraints — background only here; examinable in FEE3412.

- 2 **Transient response degrades.** Large K gives a faster but more oscillatory response, and the specification usually limits overshoot.

Weeks 5, 8, 9.

- 3 **The system can go unstable.** Above some gain the closed loop never settles. Not in this first-order model — but the norm once the plant has real dynamics.

Weeks 7, 8, 10, 11, 12 — the largest single theme of the unit.

Ever heard a PA system squeal when the volume is turned up? That is instability caused by high-gain feedback.

Feedback is not free

Common pitfall

“Closed loop is better” is not a theorem.

- It adds a sensor — cost, noise, one more thing to fail.
- It can destabilise a plant that was perfectly well behaved open loop.

It is a **trade-off**. The rest of this unit is the machinery for making that trade deliberately rather than by luck.

And a second one: error vs actuating signal

The signal leaving the first summing junction is the **actuating signal**. It equals the true error $r - y$ *only* when both transducers have unit gain. We return to this in Week 6 under *non-unity feedback*.

Design objectives and the design process

What a well-designed control system must do

Requirement	Where we deal with it
Must be stable	Weeks 7 (Routh–Hurwitz), 8 (root locus), 11–12 (Nyquist)
Must track the reference with small steady-state error	Week 6 (static error constants, system type)
Transient response must be acceptable	Weeks 5, 8–9
Disturbance and noise effects small	<i>Background only here; FEE3412</i>
Robust to model uncertainty	Gain and phase margins, Weeks 10, 12; properly in FEE3412
Respect constraints on the control input	<i>Background only here; FEE3412</i>

Khalil, p. 6.

The three that FEE3411 owns

Key idea

Stability, **steady-state accuracy** and **transient response** are the three things this unit teaches you to *analyse*.

FEE3412 teaches you to *design* for them.

If you can say which of the three a question is about, you can usually say which method it wants.

Where the mathematics comes from

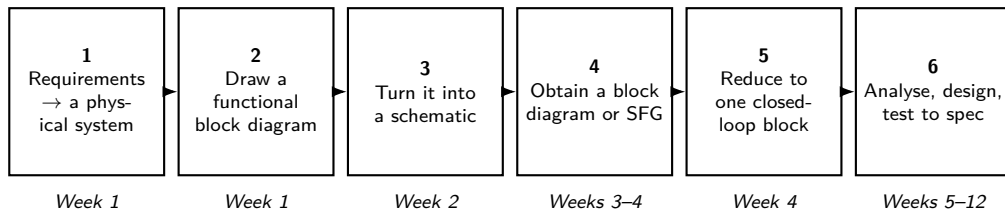
Every method in this unit needs a mathematical model of the plant. Models come from:

- **first principles** — Kirchhoff's laws, Newton's laws, energy balances. *That is Week 2.*
- **identification experiments** — apply known inputs, measure outputs, fit a model.
Outside this syllabus — but fitting a second-order model to a measured step response in Week 5 is identification in miniature.
- **both** — structure from physics, parameters by measurement.

The hard part

A model that captures everything is too complicated to design with. A model that captures too little gives a design that fails on the real plant. Choosing what to leave out requires understanding the physics, not just the algebra.

The design process — and a map of this course



Cf. Nise Fig. 1.11, p. 15. Steps 1 and 2 are today. When a topic later feels disconnected, find it on this line.

The servomechanism

The servomechanism

Definition

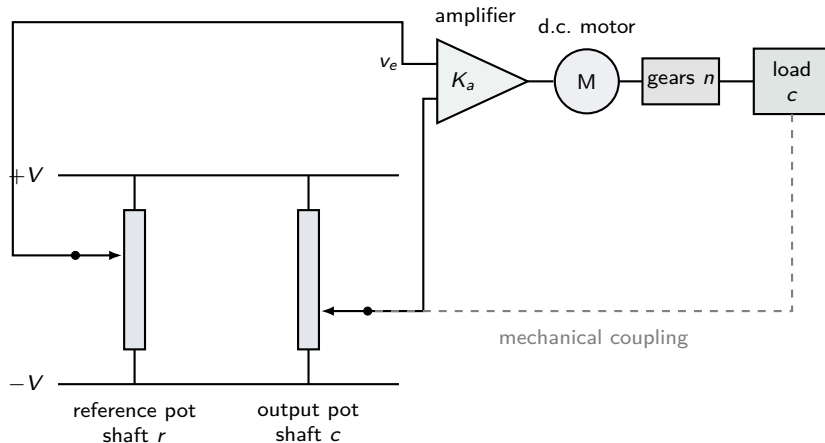
A **servomechanism** (servo) is a feedback control system in which the controlled variable is a **mechanical position**, or one of its derivatives — velocity, acceleration.

It earns its own name because it is the archetype:

- antenna pointing
- machine-tool slides
- robot joints
- radar dishes
- aircraft control surfaces
- disc-drive head positioning
- ship steering gear

Almost every worked example in this unit is, underneath, a servo.

A position servomechanism



The two wipers **are** the summing junction: $v_e = K_p(r - c)$, with K_p in V rad^{-1} . The subtraction is done by the wiring, not by a circuit. Cf. Nagrath §5.4, p. 137 (Fig. 5.6, p. 138).

Regulator or tracking system?

Regulator

Holds the output at a **constant** reference against disturbances.

Speed governor, voltage regulator, temperature control.

Tracking (follow-up)

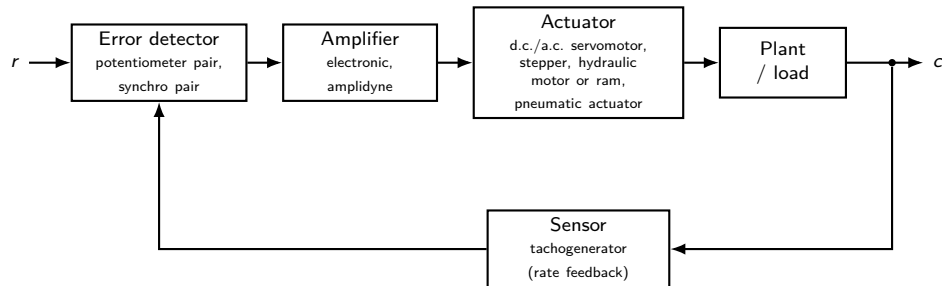
Makes the output follow a reference that is itself **changing**.

Radar dish following an aircraft, machine-tool axis following a profile.

The same hardware often does both. The distinction matters in **Week 6**, where the steady-state error depends on whether the reference is a step, a ramp or a parabola.

Components: where they sit in the loop

The four families



Depth expected

ELO 1 — *describe the working principles*. Know what a device does, how it does it, and roughly what its transfer function looks like. Deriving those transfer functions is Week 2 and, in detail, FEE3412.

Error detectors

The potentiometer pair

Two potentiometers on a common supply; one turned by the reference shaft, one by the load shaft. The voltage between the wipers is

$$v_e = K_p (r - c), \quad [K_p] = \text{V rad}^{-1}.$$

Virtues

- Cheap.
- A d.c. device — output is a d.c. voltage proportional to error.
- Model is a **pure gain** K_p , no dynamics.

Vices

- Wiper friction.
- Finite resolution — set by the winding pitch.
- Wear at the sliding contact.
- Electrical noise from that contact.

The synchro transmitter

For anything that runs continuously, turns through many revolutions, or lives in a hostile environment, a sliding contact is unacceptable. The **synchro** (*selsyn*, *autosyn*) replaces it with transformer action.

Construction — like a small three-phase alternator:

- stator: three identical Y-connected coils, axes 120° apart;
- rotor: a single concentric coil, fed a.c. through slip rings.

Applying $v(t) = V_r \sin \omega_c t$ to the rotor, the coil-to-neutral voltages are

$$v_{s_1 n} = KV_r \sin \omega_c t \cos(\theta + 120^\circ), \quad v_{s_2 n} = KV_r \sin \omega_c t \cos \theta,$$

$$v_{s_3 n} = KV_r \sin \omega_c t \cos(\theta + 240^\circ).$$

Nagrath §4.3, p. 92.

The synchro transmitter

Key idea

The synchro transmitter is a **single-phase transformer** whose rotor coil is the primary and whose three stator coils are the secondaries.

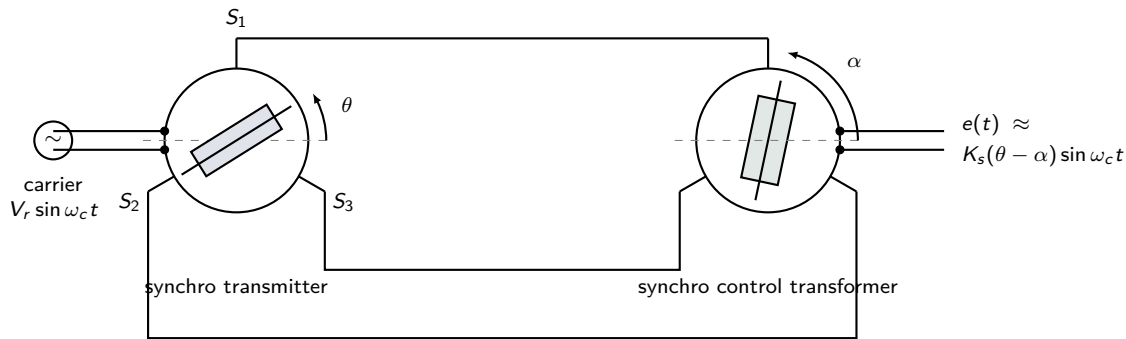
Its input is a **shaft angle**. Its output is three a.c. voltages whose magnitudes encode that angle.

The three stator terminals feed the stator of a **synchro control transformer** — built the same way, but with a *cylindrical* rotor so the air gap, and hence the rotor output impedance, does not change as it turns.

Circulating currents reproduce the transmitter's flux pattern in the control transformer's air gap, so

$$e(t) = K' V_r \cos \phi \sin \omega_c t, \quad \phi = \text{angle between the two rotors.}$$

The synchro pair as an error detector



K_s is the **sensitivity of the error detector**, V rad^{-1} . No sliding contact anywhere in the signal path. Cf. Nagrath Fig. 4.12, p. 94.

Suppressed-carrier modulation

The output is a carrier $\sin \omega_c t$ whose **amplitude** carries the error and whose **phase reverses** when the error changes sign.

A system whose internal signals look like this is a **carrier control system**. It is analysed on the modulating signal alone — valid provided the signal bandwidth is far below the carrier frequency.

Common pitfall

In a carrier system the useful information is the **envelope**, not the instantaneous value.

A student who reads the error straight off an oscilloscope trace of $e(t)$, instead of its envelope, will conclude the error is oscillating at 400 Hz while the shaft is standing perfectly still.

Electric actuators

The d.c. servomotor

The workhorse. A d.c. motor becomes a servomotor when built for low inertia and rapid reversal.

Armature control

Field current constant, armature voltage varied.

Used almost universally. Derived from first principles in the Week 2 tutorial.

Field control

Armature current constant, field voltage varied.

Cheaper amplifier, but slower — the field circuit has a large inductance.

Driving inertia J and viscous friction f :

$$\frac{\theta_m(s)}{V_a(s)} = \frac{K_m}{s(\tau_m s + 1)}$$

Note the shape — an integrator in series with a first-order lag. Keep it in mind.

The a.c. (two-phase) servomotor

Preferred at low power, and wherever a synchro error detector has already made the signals a.c. Rugged, light, no brushes.

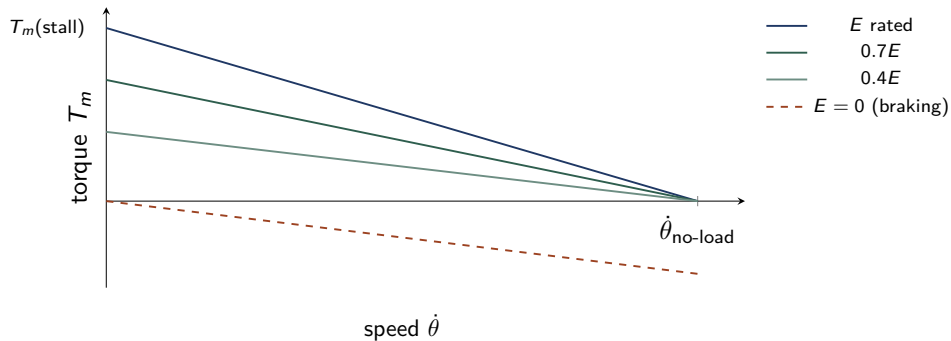
Structurally a **two-phase induction motor**: two stator windings 90° apart in space, excited 90° apart in time $\Rightarrow$ a rotating field of constant magnitude, which induces rotor currents and produces torque in the direction of field rotation.

Two design changes make it a servomotor:

- ① **High rotor resistance**, so X/R is *small*.
- ② **Unbalanced excitation** — a fixed *reference phase* and a variable *control phase* at $\pm 90^\circ$ to it. Reversing the sign of the control voltage reverses rotation. Rotor is a small-diameter squirrel cage or a drag cup, for low inertia.

Nagrath §4.3, p. 84.

Why high rotor resistance?



In an *ordinary* induction motor X/R is large, which puts peak torque near the operating point and makes part of the curve **positively** sloped. Cf. Nagrath Fig. 4.6, p. 86.

Linearising the a.c. servomotor

$T_m = f(\dot{\theta}, E)$. Expand in a Taylor series about an operating point, keep first-order terms — the technique of Week 2:

$$\Delta T_m = K \Delta E - f \Delta \dot{\theta}, \quad K = \left. \frac{\partial T_m}{\partial E} \right|_0, \quad f = - \left. \frac{\partial T_m}{\partial \dot{\theta}} \right|_0.$$

With a load of inertia J and friction f_0 :

$$G_m(s) = \frac{\Delta \theta(s)}{\Delta E(s)} = \frac{K}{Js^2 + (f_0 + f)s} = \frac{K_m}{s(\tau_m s + 1)}$$

Key idea

The **negative slope** of the torque–speed characteristic acts as extra viscous friction f , added to the load's own f_0 . If the slope were *positive*, $f_0 + f$ could go negative — negative damping — and the system would be unstable. **That is why an ordinary induction motor cannot be used as a servomotor.**

The stepper motor

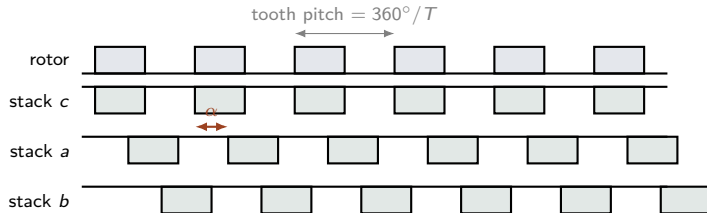
A **stepper** converts a train of input pulses into discrete angular steps — **one step per pulse**.
The actuator of incremental motion control: printers, tape and capstan drives, machine tools, plotters, 3-D printers.

In a *variable-reluctance* stepper the toothed stator stacks are pulse excited and the toothed rotors are unexcited. Energise a stack and the rotor is pulled to the nearest **minimum-reluctance** position, where the teeth line up.

- Teeth aligned $\Rightarrow$ **stable**.
- Rotor teeth facing stator slots $\Rightarrow$ **unstable**.

Nagrath §4.4, p. 99.

Stepper motor — how the step happens



$$\alpha = \frac{360^\circ}{n T} \quad \begin{array}{l} n = \text{stacks (} = \text{phases)} \\ T = \text{rotor teeth} \end{array} \quad n = 3, T = 12 \Rightarrow \alpha = 10^\circ$$

abcabc ... steps one way, *bacbac* ... the other. **Directional control needs three or more phases.** Cf. Nagrath Figs. 4.18, 4.20, pp. 100–101.

Stepper motors: the trade

A digital actuator

Shaft position is determined **entirely by the pulse count**. So an *open-loop* step servo can reach the accuracy of a closed-loop analogue system, with no position sensor at all.

This is the rare case where open loop is the right engineering answer.

The price: skipped steps

If pulses arrive faster than the rotor can settle, or the rotor oscillates too much about the lock position, a step is lost — and, since nothing is measured, lost **silently and permanently**.

High-performance systems close the loop after all, gating the pulse train from a position feedback signal.

The dynamics are genuinely nonlinear ($T_m = -K \dot{\theta}^2 \sin T\theta$) and are solved numerically — one reason steppers do not reappear in the analysis weeks.

Sensors

The tachogenerator

Produces a voltage proportional to shaft speed:

$$v_t = K_t \dot{\theta}, \quad [K_t] = \text{V rad}^{-1} \text{s}^{-1}.$$

d.c. tachogenerator

A small permanent-magnet d.c. generator. Output polarity follows the direction of rotation.

Main limitation: commutation ripple.

a.c. (drag-cup)

Two stator coils in space quadrature; rotor is a thin aluminium cup in the air gap of a fixed magnetic structure.

Advantage: very low rotor inertia.

Nagrath §4.3, p. 88.

Two quite different uses

- 1 As the **sensor of a speed control loop** — speed is the controlled variable.
- 2 As **rate feedback** inside a *position* loop — the tachometer signal is subtracted in an inner loop to add damping.

Nagrath's a.c. position control system (§4.3, p. 96; Fig. 4.14, p. 97) uses a synchro pair, an a.c. amplifier, an a.c. servomotor, gearing *and* an a.c. tachometer:

$$\frac{\theta_c(s)}{\theta_r(s)} = \frac{K_m K_a K_s n}{\tau_m s^2 + (1 + K_m K_a K_t) s + K_m K_a K_s n}$$

Key idea

K_t appears **only** in the s coefficient — the damping term. Rate feedback improves damping *without* changing the steady-state gain.

That is your first sight of a compensator. We use it as a design tool in Week 9; FEE3412 studies it properly.

Hydraulic power elements

Why hydraulics?

For

- A hydraulic motor is far smaller than an electric motor of the same power output.
- Fast acting and rugged.
- Very stiff under load.

Against

- Leaks, and sealing against contamination.
- Operating noise.
- Sluggish when cold — the oil thickens.
- Lines are not as flexible as cables.

Typical uses: power steering and brakes, ship steering gear, large machine tools, aircraft flight controls, excavators.

Two output types: *rotary* — hydraulic motors (what the syllabus calls “rotary actuators”); *translational* — linear actuators, i.e. cylinders and rams.

The pump–motor transmission

A **variable-stroke pump** driving a **fixed-stroke motor**. Both are axial-piston machines: pistons in a rotating cylinder block bear on a wobble plate. Tilt the plate through stroke angle x and oil is delivered in proportion to x ; reverse the tilt and the motor reverses.

$$q_p = K_p x \quad \text{ideal pump flow}$$

$$q_m = K_m \dot{\theta} \quad \text{motor flow}$$

$$q_\ell = K_\ell p \quad \text{leakage}$$

$$q_c = K_c \dot{p} \quad \text{compressibility}$$

$$q_p = q_m + q_\ell + q_c \quad \text{continuity}$$

$$T_m = K_T p = J \ddot{\theta} + f \dot{\theta} \quad \text{torque balance}$$

Eliminating p and neglecting compressibility ($K_c \ll K_m$):

$$G(s) = \frac{\theta(s)}{X(s)} = \frac{K}{s(\tau s + 1)} \quad \text{— the same shape again.}$$

Valve control

Instead of varying the pump stroke: hold the supply pressure constant and throttle the flow with a **spool valve**.

Gain

The moving part of a valve is far lighter than a pump's stroke mechanism, so time constants are much smaller and the system is much faster.

Cost

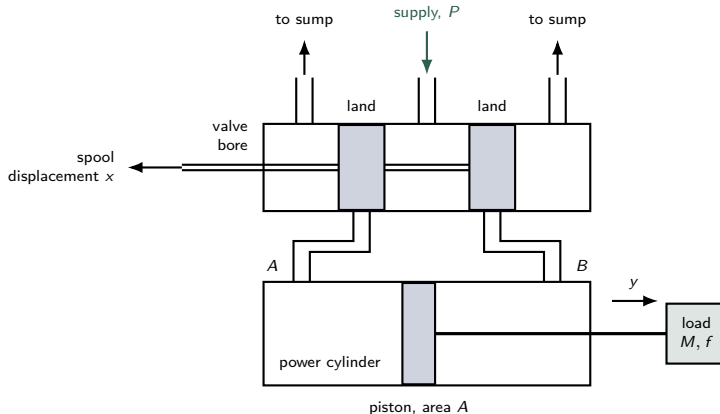
Valve flow is genuinely nonlinear — flow through a sharp-edged orifice goes as $\sqrt{\Delta p}$.

So

$$q = K_1 x - K_2 p$$

is a *linearisation*, valid only for small spool displacements about neutral. Week 2 shows the technique that produces K_1 and K_2 .

Spool valve and power cylinder



Move the spool right: B opens to supply, A drains to sump, piston is driven left. Move it the other way and the connections swap. Cf. Nagrath Fig. 4.27, p. 111.

The linear actuator (cylinder)

Three equations:

- the linearised valve relation $q = K_1 x - K_2 p$
- continuity — flow in equals swept volume rate $q = A\dot{y}$
- force balance on the load $A p = M\ddot{y} + f\dot{y}$

Eliminating q and p :

$$\frac{Y(s)}{X(s)} = \frac{K}{s(\tau s + 1)}, \quad K = \frac{K_1}{A}$$

Common pitfall — where the $1/s$ comes from

It is **not** an approximation, and it is not the motor's inertia.

Flow into a cylinder sets the piston's *velocity*, so position is the integral of the input. **Any actuator whose input commands a rate** — valve opening $\rightarrow$ velocity, armature voltage $\rightarrow$ speed, pump stroke $\rightarrow$ motor speed — carries a free integrator when the output of interest is position.

Pneumatic power elements

Air instead of oil

Advantages

- Non-inflammable — intrinsically safe.
- Negligible viscosity.
- Viscosity does not swing with temperature the way hydraulic oil does.

The catch

Air is **compressible**.

So there is substantial compressibility flow, and pneumatic systems are characterised by **longer time delays**.

Which is exactly why they dominate in *process* control — where the plant is slow anyway — and are rare where speed matters.

Nagrath §4.6, p. 116.

Bellows and the flapper-nozzle valve

Bellows — a pure gain

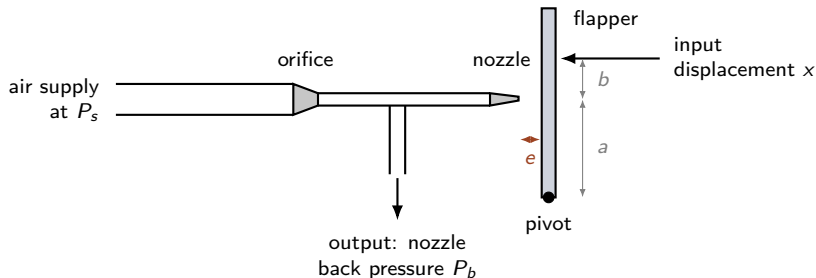
A hollow chamber with thin corrugated side walls; behaves as a spring. Separating force $= (\Delta P)A$, restoring force $= K(\Delta x)$, so

$$\frac{\Delta X(s)}{\Delta P(s)} = \frac{A}{K}.$$

The flapper-nozzle valve is the key pneumatic sensing element. Air at constant supply pressure passes through a fixed *orifice* and out of a *nozzle*; a pivoted *flapper* sits at distance e from it.

- Flapper closer $\Rightarrow$ restriction up $\Rightarrow$ back pressure P_b rises towards P_s .
- Flapper away $\Rightarrow P_b$ falls towards ambient.

The flapper-nozzle valve



$$\frac{\Delta P_b(s)}{\Delta X(s)} = \left(\frac{a}{a+b} \right) K, \quad K < 0, \quad e = \frac{a}{a+b} x$$

Small flapper movements, large pressure changes — a high-gain displacement-to-pressure transducer. Cf. Nagrath Fig. 4.34, p. 117.

The pneumatic relay

Keeping the flapper motion small enough to stay linear keeps the output pressure swing small too — so a pneumatic power amplifier, the **relay**, is put in cascade.

A ball on the lower bellows surface seats either:

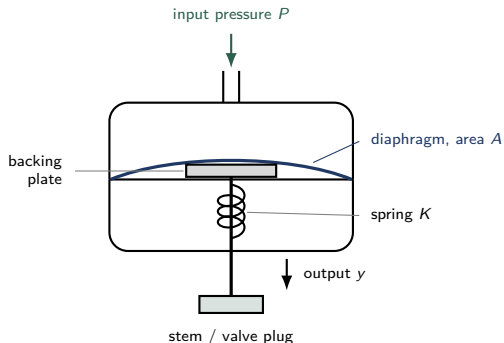
- on its **upper** seat — vent closed, output pressure rises to supply; or
- on its **lower** seat — supply shut off, output falls to ambient.

Move the flapper *away* from the nozzle $\Rightarrow P_b$ drops $\Rightarrow$ bellows contracts $\Rightarrow$ ball moves up $\Rightarrow$ vent closes $\Rightarrow$ output pressure *rises*.

So the relay inverts the sign as well as amplifying

$$\frac{\Delta P(s)}{\Delta X(s)} = \left(\frac{a}{a+b} \right) K, \quad K > 0$$

The pneumatic (diaphragm) actuator



Diaphragm of area A moves a stem against a spring K ; load contributes M and f :

$$A \Delta P = M \Delta \ddot{y} + f \Delta \dot{y} + K \Delta y$$

$$\frac{\Delta Y(s)}{\Delta P(s)} = \frac{A}{Ms^2 + fs + K}$$

Common pitfall

The hydraulic cylinder had a free $1/s$. This does **not**. The difference is the **spring**: with one, a constant pressure gives a constant *position*; without one, a constant flow gives a constant *velocity*. Read the physics, not the picture.

Electric, hydraulic, pneumatic

	Electric	Hydraulic	Pneumatic
Medium	electric current	incompressible oil	compressible air
Power/weight	moderate	very high	low
Speed	fast	fastest under load	slow (compressibility)
Stiffness	moderate	very high	low
Typical use	instruments, servos, drives	machine tools, steering, aircraft controls, presses	process control valves
Drawback	limited torque density	leaks, sealing, contamination, noise	long time delays, low stiffness
Safety	sparks	fire risk from oil mist	non-inflammable, intrinsically safe

The shape that keeps recurring

Collect the transfer functions

Device	Transfer function
Armature-controlled d.c. motor	$\theta/V_a = K_m/[s(\tau_m s + 1)]$
Two-phase a.c. servomotor	$\theta/E = K_m/[s(\tau_m s + 1)]$
Hydraulic pump-motor transmission	$\theta/X = K/[s(\tau s + 1)]$
Hydraulic linear actuator	$Y/X = K/[s(\tau s + 1)]$
Potentiometer error detector	$V_e/(r - c) = K_p$
Synchro error detector	$E/(\theta - \alpha) = K_s$
Tachogenerator	$V_t/\dot{\theta} = K_t$
Pneumatic bellows	$\Delta X/\Delta P = A/K$
Pneumatic diaphragm actuator	$\Delta Y/\Delta P = A/(Ms^2 + fs + K)$

One shape

Key idea

Almost every **positional actuator** in control engineering, whatever its physics, has the transfer function

$$G(s) = \frac{K}{s(\tau s + 1)}.$$

The $1/s$ is there because the input commands a *rate* and the output of interest is the integral of that rate. The $1/(\tau s + 1)$ is there because inertia and friction resist that rate.

Sensors and error detectors, by contrast, are usually **pure gains**.

Why that matters for the next eleven weeks

- Put $K/[s(\tau s + 1)]$ in a unity-feedback loop and you get a **second-order system** — *Week 5*, and its ζ and ω_n .
- The free integrator makes the open loop **type 1**: zero steady-state error to a step, finite error to a ramp — *Week 6*.
- Its pole–zero pattern (one pole at the origin, one on the negative real axis) gives the most common **root locus** you will ever sketch — *Week 8*.
- Its **Bode plot** is the sum of a -20 dB/decade line and one corner — *Week 10*.

Key idea

This is not a coincidence to be noted and forgotten. It is why the same handful of standard forms carry the whole unit.

Summary and what comes next

Summary — the ideas

- A control system makes a variable behave in a desired way without continuous human intervention. Parts: **plant**, **actuator**, **sensor**, **controller**.
- Open-loop applies a precomputed input and measures nothing; closed-loop measures and corrects. **Only closed-loop can reject a disturbance.**
- Feedback with gain K replaced b by $b + aK$ in the cruise-control steady-state error — shrinking the parameter-error term and the disturbance term together.
- Gain cannot be raised without limit: actuators saturate, transient response degrades, the loop may go unstable.
- A **servomechanism** is a feedback system whose controlled variable is mechanical position or a derivative of it.

Summary — the hardware

- **Error detectors:** potentiometer pair (d.c., gain K_p , sliding contact) and synchro pair (a.c., gain K_s , no sliding contact, suppressed-carrier output).
- **Actuators:** d.c. servomotor; two-phase a.c. servomotor (high rotor resistance $\Rightarrow$ negatively sloped, near-linear torque–speed); stepper (digital, step $360^\circ/nT$, usable open loop but can skip steps); hydraulic pump–motor and valve-plus-cylinder; pneumatic diaphragm actuator.
- **Sensors:** tachogenerator, $v_t = K_t \dot{\theta}$ — speed sensor *or* rate feedback for damping.
- Hydraulics: high power density and stiffness. Pneumatics: safe and cheap, but slow because air is compressible.
- Positional actuators are almost universally $K/[s(\tau s + 1)]$; error detectors and sensors are usually pure gains.

Before the tutorial

The tutorial hour is a review of the **mathematical toolkit** — Laplace pairs and properties, partial fractions, complex numbers and the s -plane, standard test signals. Revise Khalil Appendix A (p. 438) or Nise §2.2 (p. 35) beforehand.

You should be able to do these before you walk in:

- 1 A pressure cooker holds pressure with a weighted valve that lifts above a set pressure. Open or closed loop? Identify plant, controlled variable, sensor, actuator, reference, disturbance.
- 2 With $a = 20 \text{ N } ^\circ\text{ }^{-1}$, $b = 50 \text{ N s m}^{-1}$, $\hat{b} = 45 \text{ N s m}^{-1}$, $W = 200 \text{ N}$, $v_{des} = 25 \text{ m s}^{-1}$: find the steady-state error open loop, and closed loop with $K = 10$. By what factor does feedback improve it?
- 3 A three-stack VR stepper has 20 rotor teeth. Step angle? Pulses per revolution? Adequate for 0.5° positioning?

Looking ahead

- **Week 2** — models from first principles: electrical networks, translational and rotational mechanical systems. The tutorial builds the armature-controlled d.c. servomotor with a gear train — the transfer function quoted today without proof.
- **Week 3** — those models become transfer functions; into the s -plane.
- **Week 4** — block diagrams and their reduction.

Two boundaries, stated plainly

Components are survey depth here. Describe working principles (ELO 1); you are not expected to design a servomotor or select a valve. FEE3412 has a full hardware block.

FEE3411 owns analysis; FEE3412 owns design. Choosing a compensator — lead, lag, lag-lead, PID — belongs to FEE3412. When you meet “design a compensator” in Khalil Ch. 5 or Nise Ch. 9 and 11, treat it as a forward reference: worth reading, not examinable here.

Reading for this week

Primary

Khalil, Ch. 1 (pp. 1–8). Appendix A, Laplace review (p. 438), for the tutorial.

Secondary

Nise (6th ed.), §1.1–1.4 (pp. 2–14); §1.5 The Design Process (p. 15); §2.2 Laplace review (p. 35).

Components — essential this week

Nagrath & Gopal (2nd ed.), §4.3 Electrical Systems (pp. 84–99); §4.4 Stepper Motor (pp. 99–104); §4.5 Hydraulic Systems (pp. 104–116); §4.6 Pneumatic Systems (pp. 116–121); §5.4 (p. 137) for the potentiometer servomechanism.